TE 456 - Satellite Communication & NTN

200 Multiple-Choice Questions with Answers

Part 1 - Fundamentals, Orbits & Beams, 5G NR-NTN Architecture

1. A Non-Terrestrial Network (NTN) is best defined as a wireless communication network that:

A. Uses radio network equipment carried on airborne or spaceborne vehicles ✓
B. Relies on ground-based cell towers linked by buried fibre-optic backhaul
C. Connects two mobile handsets directly via a sidelink, bypassing any base station
D. Uses free-space optical laser links between two fixed ground stations on Earth

Answer: A. Uses radio network equipment carried on airborne or spaceborne vehicles

2. Terrestrial networks (TN) and non-terrestrial networks (NTN) differ primarily in their:

A. Infrastructure and coverage areas ✓
B. Choice of programming language for the core network
C. Use of the TCP/IP protocol suite
D. Billing and subscription models

Answer: A. Infrastructure and coverage areas

3. In NTN terminology, a 'spaceborne' platform refers to:

A. Drones and stratospheric balloons flying in the lower and upper atmosphere
B. Fibre-optic backbone cabling that links terrestrial data centres together
C. Satellite communication using artificial satellites orbiting the Earth ✓
D. Ground gateway stations that steer directional antennas toward orbiting satellites

Answer: C. Satellite communication using artificial satellites orbiting the Earth

4. Which of the following are examples of airborne NTN platforms?

A. GEO, MEO and LEO satellites orbiting hundreds to tens of thousands of km up
B. Fixed terrestrial gNBs connected by underground fibre-optic cables
C. Undersea fibre cables and ground-based microwave relay towers
D. Drones (UAVs) and High-Altitude Platform Stations (HAPS) ✓

Answer: D. Drones (UAVs) and High-Altitude Platform Stations (HAPS)

5. A High-Altitude Platform Station (HAPS), such as a balloon or airship in the stratosphere, is best characterised as an NTN platform that is:

A. A navigation satellite in medium Earth orbit at an altitude of about 20,000 km
B. Unmanned, with lower latency than LEO and easier maintainability than a satellite ✓
C. A ground-based mast and antenna forming a fixed part of the terrestrial network
D. A geostationary spacecraft fixed in orbit above a point on the equator

Answer: B. Unmanned, with lower latency than LEO and easier maintainability than a satellite

6. One motivation for NTN cited in the 6G vision is that terrestrial networks currently:

A. Cover less than 40% of the Earth's surface ✓
B. Cover more than 95% of the Earth's surface
C. Already provide 5G coverage across more than 90% of the Earth's landmass
D. Provide uniform, seamless coverage over the oceans

Answer: A. Cover less than 40% of the Earth's surface

7. An NTN platform creating large beams to illuminate remote farms, oceans and IoT devices anywhere on Earth illustrates which NTN use-case category?

A. Service continuity
B. Service scalability
C. Service ubiquity ✓

D. Service localization

Answer: C. Service ubiquity

8. A cellular subscriber who boards a cruise ship or an aircraft and would otherwise lose coverage is best served by which NTN use-case category?

A. Service ubiquity
B. Service scalability
C. Service redundancy
D. Service continuity ✓

Answer: D. Service continuity

9. Broadcasting the same content, such as a software update or a live event, to many users over a large area with one wide NTN beam illustrates:

A. Service scalability ✓
B. Service continuity
C. Service ubiquity
D. Service isolation

Answer: A. Service scalability

10. Deploying terrestrial base stations across sparsely populated deserts, mountains and oceans is generally avoided mainly because:

A. Radio waves cannot physically travel over open water
B. Spectrum is legally unavailable outside cities
C. Satellites legally prohibit ground towers in those areas
D. It is not cost-effective and often lacks a viable business case ✓

Answer: D. It is not cost-effective and often lacks a viable business case

11. A major driver of the decades-long success of terrestrial mobile networks has been:

A. The complete absence of any government regulation across the sector
B. International standardization ensuring vendor compatibility and lower costs ✓
C. Exclusive reliance on proprietary, vendor-locked, closed hardware platforms
D. The use of a single global mobile operator that dominates the worldwide market

Answer: B. International standardization ensuring vendor compatibility and lower costs

12. Historically, the satellite communications market has been fragmented largely because of:

A. Excessive standardization efforts that limited vendor innovation and design freedom
B. Government bans on the domestic manufacturing of communication satellites
C. An abundant supply of low-cost, interchangeable satellite devices
D. A lack of standardization, making interoperability between vendors difficult ✓

Answer: D. A lack of standardization, making interoperability between vendors difficult

13. In a TN-versus-NTN comparison, the typical cell size differs in that:

A. TN cells are typically massive in size, whereas NTN cells shrink to just a few metres across in dense deployments
B. TN cells are small (giving high capacity per cell), whereas NTN cells are massive, tens to thousands of km across ✓
C. Both TN and NTN use cells of an identical, fixed diameter regardless of platform or altitude
D. NTN cells are consistently smaller than TN cells specifically to limit inter-cell interference levels

Answer: B. TN cells are small (giving high capacity per cell), whereas NTN cells are massive, tens to thousands of km across

14. The infrastructure of a terrestrial network is characterised by:

A. Satellites, HAPS platforms and drones operating high above the ground
B. End-user handsets communicating directly with each other via sidelink, peer-to-peer
C. Fixed ground-based base stations (e.g., gNodeB), cables and switching centres ✓
D. Optical inter-satellite links relaying traffic between spacecraft in orbit

Answer: C. Fixed ground-based base stations (e.g., gNodeB), cables and switching centres

15. The current relationship between the satellite and terrestrial mobile communication industries is best described as:

A. Progressively converging toward TN-NTN integration ✓
B. Diverging into two fully separate, non-interoperable ecosystems
C. Remaining two historically isolated industrial supply chains
D. Being steadily replaced by fixed, wired fibre-optic networks

Answer: A. Progressively converging toward TN-NTN integration

16. The 3rd Generation Partnership Project (3GPP) is significant to NTN because it:

A. Manufactures and launches its own fleet of communication satellites into orbit
B. Completed the first global 5G NR standard and is extending it to support satellites ✓
C. Owns and commercially operates the Starlink broadband satellite constellation
D. Issues national spectrum licences to mobile operators in each member country

Answer: B. Completed the first global 5G NR standard and is extending it to support satellites

17. In 5G rollout, NTN is expected to serve unserved areas such as isolated regions, aircraft and vessels, and additionally to serve:

A. Underserved areas such as sub-urban/rural regions, upgrading limited terrestrial performance ✓
B. Dense urban centres that already enjoy full 5G coverage from terrestrial gNodeBs
C. Areas that already have a wired fibre-optic backhaul connection installed nearby
D. Military installations under a dedicated defence spectrum allocation, apart from civilian networks

Answer: A. Underserved areas such as sub-urban/rural regions, upgrading limited terrestrial performance

18. Generally, compared with NTN, terrestrial networks are described as providing:

A. Lower reliability and a higher bit error rate under typical conditions
B. Identical reliability regardless of weather or terrain conditions
C. Higher reliability and lower bit error rates ✓
D. Reliable service specifically over open-ocean shipping routes

Answer: C. Higher reliability and lower bit error rates

19. A frequently cited advantage of NTN over TN during earthquakes, floods or armed conflict is that NTN:

A. Can maintain coverage when terrestrial infrastructure is damaged or destroyed ✓
B. Requires the same fibre-based backhaul infrastructure as terrestrial networks do
C. Relies on a single regenerative satellite payload rather than any ground gateway
D. Operates over dedicated fibre-optic links instead of licensed radio spectrum

Answer: A. Can maintain coverage when terrestrial infrastructure is damaged or destroyed

20. Comparing latency, terrestrial networks are typically ultra-low (under about 1 ms), whereas non-terrestrial networks range from:

A. About 20 ms for LEO up to 500 ms or more for GEO, owing to the large distances ✓
B. About 1 ms for LEO up to roughly 5 ms for GEO, similar to fibre backhaul
C. A fixed 100 ms round-trip delay regardless of which orbit type is used
D. Consistently lower latency than terrestrial fibre networks in most deployment cases

Answer: A. About 20 ms for LEO up to 500 ms or more for GEO, owing to the large distances

21. Which statement best summarises the primary limitations of terrestrial networks?

A. Effectively unlimited global reach but consistently very high transmission latency
B. Restricted global reach, vulnerability to physical damage, and high cost in rugged terrain ✓
C. Excellent resilience to natural disasters but comparatively poor data throughput
D. Strong global ocean coverage but comparatively weak coverage in dense urban centres

Answer: B. Restricted global reach, vulnerability to physical damage, and high cost in rugged terrain

22. Regarding mobility support, a key contrast between the two networks is that:

A. A TN requires frequent handovers, while an NTN can offer seamless mobility across wide areas ✓

B. A TN offers seamless wide-area mobility, while an NTN needs frequent, active handovers instead

C. Neither a TN nor an NTN is designed to support user mobility in current standards

D. Both require one scheduled handover procedure at a fixed daily interval

Answer: A. A TN requires frequent handovers, while an NTN can offer seamless mobility across wide areas

23. Relative to extensive terrestrial deployments in remote regions, NTNs are described as:

A. Consistently more expensive than terrestrial builds, with no added coverage benefit

B. Roughly equivalent in total deployment cost but offering lower area coverage

C. Potentially more cost-effective, while extending coverage and providing a backup ✓

D. Cheaper mainly because satellite operators are exempt from national spectrum licensing fees

Answer: C. Potentially more cost-effective, while extending coverage and providing a backup

24. In the Ghanaian context discussed, rural regions have largely lagged in 4G coverage, leaving many users dependent on:

A. 5G standalone networks

B. Satellite broadband that is already deployed

C. Fibre-to-the-home connections

D. 3G networks ✓

Answer: D. 3G networks

25. Compared with a terrestrial network's standard path loss, the signal constraints of an NTN are dominated by:

A. Minimal path loss and effectively no Doppler shift, resembling a short-range terrestrial Wi-Fi link

B. Interference arising mainly from neighbouring ground-based cellular towers and street-level clutter

C. A near-complete absence of any meaningful propagation loss, fading, or spectral interference effects

D. Severe path loss and high Doppler shifts from satellite mobility, plus atmospheric/ionospheric interference ✓

Answer: D. Severe path loss and high Doppler shifts from satellite mobility, plus atmospheric/ionospheric interference

26. According to Kepler's laws as applied to satellites, satellite orbits are:

A. Perfect circles centred exactly on the Earth's core

B. Straight lines tangent to the equator

C. Squares traced around the two poles

D. Ellipses with the Earth at one of the foci ✓

Answer: D. Ellipses with the Earth at one of the foci

27. In an altitude-versus-orbital-speed table, orbital speed falls from about 7.8 km/s near 200 km to about 3.1 km/s at roughly 35,800 km. This illustrates that:

A. Orbital speed increases as altitude increases, since higher orbits require greater velocity

B. Orbital speed decreases as altitude increases, while the orbital period lengthens ✓

C. Orbital speed remains essentially constant regardless of orbital altitude

D. The orbital period shortens as altitude increases, mirroring the decrease in orbital speed

Answer: B. Orbital speed decreases as altitude increases, while the orbital period lengthens

28. A geostationary (GEO) satellite orbits at an altitude of approximately:

A. 35,786 km ✓

B. 550 km

C. 20,200 km

D. 384,000 km

Answer: A. 35,786 km

29. A GEO satellite appears stationary to a fixed observer on Earth because it:

A. Does not actually move within its orbit, remaining fixed relative to the stars

B. Completes one orbit in about 24 hours, matching the Earth's rotation ✓

C. Orbits the Earth in under 90 minutes, much faster than the planet's rotation

D. Is physically held in place by a tether anchored to a ground station below

Answer: B. Completes one orbit in about 24 hours, matching the Earth's rotation

30. Theoretically, near-global coverage (excluding the extreme poles) can be achieved with how many GEO satellites?

A. One

B. Twelve

C. Twenty-four

D. Three ✓

Answer: D. Three

31. Because of its high altitude, a GEO satellite exhibits a round-trip propagation delay on the order of:

A. About 1 ms

B. About 600 ms ✓

C. About 25 ms

D. About 50 microseconds

Answer: B. About 600 ms

32. A recognised disadvantage of GEO satellites is:

A. The need for constant satellite-to-satellite handovers

B. An extremely short, negligible propagation delay

C. A requirement for thousands of satellites for coverage

D. Poor signal coverage at the extreme polar regions ✓

Answer: D. Poor signal coverage at the extreme polar regions

33. A satellite whose orbital period matches the Earth's rotation but which may be inclined or elliptical, lacking the strict 0-degree equatorial, fixed-point constraint of a GEO, is classified as:

A. Low Earth orbit (LEO), typically a few hundred kilometres up

B. Medium Earth orbit (MEO), used mainly for navigation constellations

C. Geosynchronous orbit (GSO), of which GEO is a special case ✓

D. A High-Altitude Platform Station (HAPS) flying in the stratosphere

Answer: C. Geosynchronous orbit (GSO), of which GEO is a special case

34. Low Earth Orbit (LEO) satellites typically operate at altitudes of:

A. 300 to 1,500 km ✓

B. 7,000 to 25,000 km

C. Exactly 35,786 km

D. Above 100,000 km

Answer: A. 300 to 1,500 km

35. A typical LEO satellite has an orbital period of about:

A. 24 hours

B. 6 to 12 hours

C. 90 minutes ✓

D. One week

Answer: C. 90 minutes

36. Because of their low altitude, LEO satellites offer a small one-way propagation delay on the order of:

A. About 600 ms

B. About 83 ms
C. Several seconds
D. 1 to 5 ms ✓
Answer: D. 1 to 5 ms

37. Medium Earth Orbit (MEO) satellites orbit at altitudes ranging from about:

A. 300 to 1,500 km
B. 7,000 to 25,000 km ✓
C. 35,786 km, fixed
D. Below 300 km
Answer: B. 7,000 to 25,000 km

38. MEO satellites are described as the backbone of global navigation; for example, GPS uses at least:

A. 24 MEO satellites ✓
B. 3 GEO satellites
C. Thousands of LEO satellites
D. A single HAPS platform
Answer: A. 24 MEO satellites

39. A navigation satellite orbiting at an altitude of about 20,200 km is described as semi-synchronous, meaning its orbital period is approximately:

A. 24 hours
B. 12 hours ✓
C. 90 minutes
D. 1 hour
Answer: B. 12 hours

40. Starlink provides broadband using thousands of LEO satellites flying at about 550 km, achieving a round-trip delay of roughly:

A. 600 ms
B. 250 ms
C. 25 ms ✓
D. 2 ms
Answer: C. 25 ms

41. A disadvantage of NGEO (LEO/MEO) systems compared with GEO is that they:

A. Have significantly higher round-trip latency than a comparable GEO satellite link
B. Require dynamic hand-offs and active ground-antenna tracking of moving satellites ✓
C. Cannot provide meaningful coverage of the polar regions at high latitudes
D. Need a single stationary dish pointed at a fixed spot, with no active tracking
Answer: B. Require dynamic hand-offs and active ground-antenna tracking of moving satellites

42. For High-Altitude Platform Stations (HAPS), 3GPP focuses on altitudes between:

A. 300 and 1,500 km
B. 8 and 50 km ✓
C. 35,000 and 36,000 km
D. 1 and 5 km
Answer: B. 8 and 50 km

43. Because a HAPS maintains its position relative to the Earth's surface, it:

A. Uses Earth-moving beams that sweep across new ground area, like a fast-moving LEO satellite
B. Has a round-trip propagation delay of about 600 ms, similar to a GEO satellite link
C. Uses Earth-fixed beams like a GEO satellite, with delay comparable to terrestrial networks ✓
D. Covers a ground area smaller than that of a single terrestrial gNB cell site
Answer: C. Uses Earth-fixed beams like a GEO satellite, with delay comparable to terrestrial

networks

44. Compared with spaceborne NTN, airborne NTN platforms:

A. Have a considerably larger end-to-end propagation delay than a comparable GEO satellite link
B. Are largely unaffected by moderate winds and storms, unlike ground-based masts
C. Provide continuous global coverage from a single stationary platform overhead
D. Deploy quickly at lower cost with smaller delay, but face stabilization and weather challenges ✓

Answer: D. Deploy quickly at lower cost with smaller delay, but face stabilization and weather challenges

45. For an Unmanned Aircraft System (UAS) used as an airborne NTN platform, the drone's operating altitude is typically limited to about:

A. 8 to 50 km, up in the stratosphere where a HAPS typically operates
B. 300 to 1,500 km, the typical altitude range for a LEO satellite
C. 90 to 150 m, depending on country and region ✓
D. 35,786 km, the fixed altitude of a GEO satellite

Answer: C. 90 to 150 m, depending on country and region

46. The three types of beams used in an NTN to provide radio coverage are Earth-fixed, Earth-moving, and:

A. Polar-locked beams
B. Quasi-Earth-fixed beams ✓
C. Counter-rotating beams
D. Ground-anchored beams

Answer: B. Quasi-Earth-fixed beams

47. An Earth-fixed beam is one that:

A. Sweeps across a new patch of ground continuously as the satellite moves
B. Covers the same fixed geographic region on Earth at all times ✓
C. Covers area X, then abruptly jumps to a distant area Y without transition
D. Is generated by a fast-moving LEO satellite rather than a stationary one

Answer: B. Covers the same fixed geographic region on Earth at all times

48. An Earth-moving beam is characterised by the fact that it:

A. Continuously illuminates the exact same ground area throughout the satellite pass
B. Covers a slightly different geographic area at each successive instant as the platform moves ✓
C. Is produced by a stationary GEO satellite that does not move relative to Earth
D. Requires active beam steering to remain fixed on one ground spot continuously

Answer: B. Covers a slightly different geographic area at each successive instant as the platform moves

49. With a quasi-Earth-fixed beam covering area X from t1 to t2 and then a different area, continuity over area X is maintained because:

A. The satellite physically slows down and briefly stops moving along its orbit
B. A single wide beam covers the entire visible hemisphere simultaneously
C. The user device seamlessly switches over to a wired terrestrial connection instead
D. An incoming quasi-Earth-fixed beam replaces the outgoing beam over the same area ✓

Answer: D. An incoming quasi-Earth-fixed beam replaces the outgoing beam over the same area

50. GEO satellites, with beam footprints from roughly 200 to 3,500 km, are most commonly used for services such as:

A. Ultra-low-latency competitive online gaming requiring sub-20 ms response times
B. Sub-millisecond industrial control loops for robotic factory automation
C. Short-range indoor positioning within a single warehouse or building
D. Weather monitoring, TV broadcasting, and remote sensing/positioning ✓

Answer: D. Weather monitoring, TV broadcasting, and remote sensing/positioning

51. In 5G NR-NTN, the 'service link' is:

A. The feeder-link connection between the satellite and the ground gateway station
B. The direct radio link between two satellites relaying traffic in orbit
C. The 5G radio access (Uu interface) between the UE and the satellite ✓
D. The wired terrestrial link between the ground gateway and the 5G core network

Answer: C. The 5G radio access (Uu interface) between the UE and the satellite

52. The 'feeder link' (satellite radio interface, SRI) describes the connection between:

A. The UE and the satellite over the radio service link
B. The gNB and the 5G core network via a wired N2/N3 interface
C. The satellite and the ground gateway ✓
D. Two user devices communicating directly via sidelink

Answer: C. The satellite and the ground gateway

53. An NTN round-trip time is expressed as RTT = 2(Tprop1 + Tprop2) + 2 Tslot + Tproc1 + Tproc2. In this expression, the propagation terms Tprop1 and Tprop2 correspond to the:

A. UE-to-satellite link and the satellite-to-gateway link ✓
B. Uplink and downlink of a single terrestrial cell
C. Two processing steps carried out inside the gNB
D. Inter-satellite link and the terrestrial fibre backhaul

Answer: A. UE-to-satellite link and the satellite-to-gateway link

54. A transparent (bent-pipe) NTN payload:

A. Amplifies, filters and frequency-translates the signal without onboard baseband processing ✓
B. Fully decodes, error-corrects and re-encodes the signal onboard before retransmission
C. Hosts the complete gNB, including RRC and PDCP layers, onboard the satellite
D. Terminates the 5G Uu radio interface directly on the satellite payload itself

Answer: A. Amplifies, filters and frequency-translates the signal without onboard baseband processing

55. With a transparent payload, the NTN gNB is located:

A. Onboard the satellite itself, alongside the on-board processor
B. Inside the UE handset, integrated with the baseband modem
C. Distributed redundantly across several satellites in the same orbital plane
D. On the ground, typically co-located with the gateway ✓

Answer: D. On the ground, typically co-located with the gateway

56. In the transparent-payload architecture, the 5G Uu radio interface terminates at:

A. The satellite payload, acting as a transparent relay point
B. The terrestrial gNB, not at the satellite ✓
C. The UE alone, with no counterpart terminating on the network side
D. The 5G core network, several hops beyond the ground gateway

Answer: B. The terrestrial gNB, not at the satellite

57. A regenerative (decode-and-forward) payload differs from a transparent one in that it:

A. Simply amplifies and frequency-converts the signal, with no onboard processing
B. Removes the need for a ground-based gateway station between hops
C. Incorporates gNB functions onboard, terminating the Uu interface at the satellite ✓
D. Cannot support any inter-satellite links due to onboard hardware limits

Answer: C. Incorporates gNB functions onboard, terminating the Uu interface at the satellite

58. A key latency advantage of the regenerative architecture is that:

A. The satellite hardware becomes far simpler, cheaper, and lighter to launch

B. It removes the need for a service link between the UE and the satellite altogether
C. The single-way latency includes only the service link, reducing round-trip time ✓
D. The Uu interface terminates on the ground at the gateway, not in orbit

Answer: C. The single-way latency includes only the service link, reducing round-trip time

59. An Inter-Satellite Link (ISL) in a regenerative NTN:

A. Connects a satellite to a terrestrial fibre ring encircling a data centre
B. Connects NTN payloads to one another, allowing relay toward a distant gateway ✓
C. Links a UE directly to the 5G core network without any radio interface
D. Replaces the feeder link with a buried fibre-optic cable to the gateway

Answer: B. Connects NTN payloads to one another, allowing relay toward a distant gateway

60. A disaggregated gNB is split into a:

A. Uplink Processing Unit and a separate Downlink Processing Unit
B. Service Link Unit and a distinct Feeder Link Unit
C. Primary Radio Unit and a redundant Secondary Radio Unit
D. Centralized Unit (CU) and a Distributed Unit (DU) ✓

Answer: D. Centralized Unit (CU) and a Distributed Unit (DU)

61. When a regenerative payload hosts only the gNB-DU while the CU stays on the ground, the feeder link carries the:

A. F1 interface (F1-C signalling and F1-U user-plane traffic) ✓
B. Uu air interface signalling directly between the UE and the DU
C. N6 interface carrying user traffic toward the public Internet
D. ISL optical control channel linking neighbouring satellites

Answer: A. F1 interface (F1-C signalling and F1-U user-plane traffic)

62. In a multi-connectivity NTN scenario, the NTN UE:

A. Connects to exactly one satellite at any given time, with no fallback
B. Relies on a wired backhaul connection instead of any radio link
C. Cannot connect to any terrestrial network alongside the satellite link
D. Simultaneously communicates with multiple radio or core networks ✓

Answer: D. Simultaneously communicates with multiple radio or core networks

63. For direct NTN access to handheld UEs within FR1, the commonly used frequency bands are:

A. The mmWave FR2 bands above 24 GHz, unsuitable for handheld devices
B. Optical and visible-light bands used for free-space laser links
C. S-band and L-band, the sub-6 GHz FR1 ranges used for handheld access ✓
D. Sub-100 MHz HF bands used for long-distance ionospheric skywave links

Answer: C. S-band and L-band, the sub-6 GHz FR1 ranges used for handheld access

64. In the NTN HARQ retransmission process, after the receiver returns a negative acknowledgement (NACK) for a corrupted packet, the transmitter responds by:

A. Dropping the connection to the UE after a small fixed number of retries
B. Retransmitting the data using a new redundancy version (RV) ✓
C. Switching the UE over to a different satellite constellation mid-session
D. Ignoring the NACK and simply moving on to the next scheduled data block

Answer: B. Retransmitting the data using a new redundancy version (RV)

65. The most challenging NTN characteristic inhibiting low-latency communication is:

A. The small physical size of the satellite antenna array limiting gain
B. The long round-trip time due to the large UE-to-satellite distance ✓
C. The absence of a standardized modulation scheme for the downlink carrier
D. The relatively low deployment cost of terrestrial ground gateway stations

Answer: B. The long round-trip time due to the large UE-to-satellite distance

66. Typical one-way latency values quoted for NTN range from about 30-40 ms in LEO up to:

A. About 5 ms in GEO constellations
B. About 1 ms in GEO constellations
C. About 544 ms in GEO constellations ✓
D. About 60 ms in GEO constellations

Answer: C. About 544 ms in GEO constellations

67. The Doppler (carrier-frequency) shift in NTN arises mainly because:

A. The satellite remains perfectly stationary relative to a fixed point on the Earth's surface
B. The satellite, and possibly the UE, moves, causing a time-variant carrier-frequency deviation ✓
C. The ground gateway periodically changes its assigned IP address mid-session
D. The UE gradually increases its transmit power as the session progresses

Answer: B. The satellite, and possibly the UE, moves, causing a time-variant carrier-frequency deviation

68. The 'Doppler rate' in an NTN specifically refers to:

A. The fixed frequency offset measured at a single instant in time
B. The variation of the Doppler shift over the connection time ✓
C. The data throughput rate carried on the satellite feeder link
D. The rate at which new satellites are launched into a constellation

Answer: B. The variation of the Doppler shift over the connection time

69. To compensate the uplink Doppler shift, the UE typically:

A. Estimates its position via GNSS and uses satellite ephemeris to pre-adjust its uplink carrier frequency ✓
B. Randomly hops across the available subcarriers, hoping the receiver stays synchronized despite the shift
C. Increases its uplink transmit power in an attempt to overpower the frequency shift
D. Waits passively until the satellite pauses its motion along the orbital path

Answer: A. Estimates its position via GNSS and uses satellite ephemeris to pre-adjust its uplink carrier frequency

70. Ionospheric propagation that causes a rotation of the waveform's polarization is known as:

A. Doppler rotation
B. Keplerian precession
C. Rayleigh fading
D. Faraday rotation ✓

Answer: D. Faraday rotation

71. Beyond round-trip time and Doppler effects, a further NTN technical challenge affecting the PHY/MAC layers is:

A. Moving cells caused by satellite motion ✓
B. Perfectly static, unchanging coverage areas
C. The complete absence of any base station
D. An unlimited supply of available spectrum

Answer: A. Moving cells caused by satellite motion

72. In the transparent-payload protocol stack, the termination points for RRC, PDCP, RLC, MAC and PHY are on the:

A. Satellite payload and the ground gateway station, since both process baseband
B. NTN UE and the NTN gNB, since the satellite and gateway are transparent ✓
C. 5G core network and the public Internet, several hops beyond the gateway
D. UE and the satellite alone, without the gateway or gNB participating

Answer: B. NTN UE and the NTN gNB, since the satellite and gateway are transparent

73. When a regenerative payload houses the full gNB, the RRC terminates in the satellite, which results in:

A. A much longer RRC reaction time than the transparent case
B. No RRC layer being present anywhere in the regenerative architecture
C. RRC termination out on the Internet
D. A shorter reaction time for RRC procedures ✓

Answer: D. A shorter reaction time for RRC procedures

74. In NTN, the gateway is the terrestrial station that:

A. Acts as the end-user handset
B. Generates the satellite's onboard electrical power
C. Connects the gNB (or 5GC) functions to the satellite ✓
D. Provides GNSS positioning directly to the UE

Answer: C. Connects the gNB (or 5GC) functions to the satellite

75. The regenerative payload's on-board processor (OBP), inserted between the LNA and the HPA, enables:

A. Purely analogue amplification and frequency conversion, with no digital processing
B. Elimination of the separate downlink antenna, reusing the uplink array instead
C. Error correction and routing of packets between beams before retransmission ✓
D. A direct wired fibre connection from the satellite down to the 5G core network

Answer: C. Error correction and routing of packets between beams before retransmission

76. At its most basic, a communication satellite acts as a:

A. A passive mirror that simply reflects incoming signals unchanged, with no amplification
B. A data-storage warehouse in orbit, downloading its contents once per day
C. Repeater - a receiver linked to a transmitter using different frequencies ✓
D. A GNSS timing clock broadcasting position data, with no communication relay function

Answer: C. Repeater - a receiver linked to a transmitter using different frequencies

77. A satellite communication system is divided into two main segments:

A. The uplink segment and the billing segment
B. The analogue segment and the digital segment
C. The civilian segment and the military segment
D. The space segment and the earth/ground segment ✓

Answer: D. The space segment and the earth/ground segment

78. The communications payload of a satellite consists of two distinct parts:

A. Solar panels and batteries
B. Thrusters and fuel tanks
C. Ground gateways and user terminals
D. Repeaters/transponders and antennas ✓

Answer: D. Repeaters/transponders and antennas

79. A transponder changes the uplink carrier frequency to a different downlink frequency. In the Ku band this is, for example, from:

A. 14 GHz on the uplink to 11 GHz on the downlink ✓
B. 11 GHz on the uplink to 14 GHz on the downlink
C. 20 GHz on the uplink to 30 GHz on the downlink
D. 14 GHz on the uplink to 14 GHz on the downlink

Answer: A. 14 GHz on the uplink to 11 GHz on the downlink

80. If a payload bandwidth of 500 MHz is divided into channels of 36 MHz each, the approximate number of transponders is:

A. About 12 ✓
B. About 500

C. About 36
D. About 2

Answer: A. About 12

81. A satellite repeater always uses different frequencies for receive and transmit mainly in order to:

A. Avoid interference between the incoming and outgoing signals ✓
B. Save electrical power onboard the satellite by cutting amplifier duty cycles
C. Comply with Kepler's laws of motion governing the shape of the orbit
D. Reduce the satellite payload down to a single shared antenna element

Answer: A. Avoid interference between the incoming and outgoing signals

82. A transparent (bent-pipe) repeater typically consists of a low-noise amplifier, a high-power amplifier, and:

A. A mixer with a local oscillator for frequency conversion ✓
B. An on-board baseband processor that fully decodes and re-encodes the signal
C. A GNSS receiver used solely for determining the satellite's own position
D. A propulsion thruster used for periodic orbital station-keeping manoeuvres

Answer: A. A mixer with a local oscillator for frequency conversion

83. Unlike a transparent repeater, a regenerative (processing) repeater:

A. Simply amplifies and forwards everything it receives, including any accumulated noise
B. Demodulates the uplink to baseband, corrects errors, then remodulates for the downlink ✓
C. Cannot translate the uplink carrier frequency to a different downlink frequency
D. Contains no power amplifiers, relying purely on passive reflection of the signal

Answer: B. Demodulates the uplink to baseband, corrects errors, then remodulates for the downlink

84. A benefit of a regenerative repeater over a bent-pipe one is that it:

A. Consumes noticeably less electrical power and structural mass than a bent-pipe design
B. Operates without any onboard antenna, coupling signals directly through waveguides
C. Works with analogue signals alone, unable to process digital baseband data
D. Cleans the signal by decoding and correcting errors instead of forwarding noise ✓

Answer: D. Cleans the signal by decoding and correcting errors instead of forwarding noise

85. Which of the following is a space-platform (bus) subsystem that supports the payload?

A. Physical Downlink Shared Channel (PDSCH)
B. User Plane Function (UPF)
C. Resource Block scheduler
D. Attitude and Orbit Control System (AOCS) ✓

Answer: D. Attitude and Orbit Control System (AOCS)

86. The Telemetry, Tracking, and Command (TTC&M) subsystem is responsible for:

A. Amplifying and retransmitting the user carriers between the uplink and downlink paths
B. Allocating dynamic IP addresses to each connected user device
C. Monitoring subsystem health, tracking position, and executing commands from ground control ✓
D. Performing OFDMA resource-block scheduling for the user-plane traffic

Answer: C. Monitoring subsystem health, tracking position, and executing commands from ground control

87. The three main satellite links are the service link, the feeder link, and the:

A. Inter-satellite link (ISL) ✓
B. Fibre backhaul link
C. Ethernet control link
D. Terrestrial microwave link

Answer: A. Inter-satellite link (ISL)

88. Inter-satellite links (ISLs) can be implemented using:

A. Either radio-frequency (RF) links or optical (laser) links ✓
B. Copper coaxial cabling strung physically between adjacent satellites
C. Acoustic signalling pulses transmitted through the vacuum of space
D. Undersea fibre-optic cable strung between satellites in orbit

Answer: A. Either radio-frequency (RF) links or optical (laser) links

89. The performance of transmitting equipment is measured by its EIRP, which is:

A. The ratio of antenna receive gain to the system's equivalent noise temperature
B. The Boltzmann constant multiplied by the absolute system noise temperature
C. The power fed to the antenna multiplied by the antenna gain in the considered direction ✓
D. The received carrier power divided by the noise power spectral density

Answer: C. The power fed to the antenna multiplied by the antenna gain in the considered direction

90. The receiver figure of merit, G/T, is the ratio of:

A. The ratio of transmit power fed into the antenna to its gain
B. The antenna receive gain to the system noise temperature ✓
C. The ratio of received carrier power to the user's data bit rate
D. The ratio of the uplink carrier frequency to the downlink carrier frequency

Answer: B. The antenna receive gain to the system noise temperature

91. A satellite link budget is fundamentally used to:

A. Predict whether the received signal will be strong enough relative to the noise ✓
B. Schedule the optimal launch window for the satellite's rocket vehicle
C. Allocate telephone numbers and IP addresses to individual subscriber devices
D. Determine the orbital inclination the satellite should maintain over its lifetime

Answer: A. Predict whether the received signal will be strong enough relative to the noise

92. Noise power spectral density N0 is defined as:

A. The antenna receive gain divided by the physical system temperature in kelvin
B. The transmitted EIRP minus the total free-space path loss along the link
C. The product of the Boltzmann constant k and the system noise temperature T ✓
D. The received carrier power multiplied by the transmitted bit rate

Answer: C. The product of the Boltzmann constant k and the system noise temperature T

93. Which of the following is a link (propagation) loss accounted for in a satellite link budget?

A. The Boltzmann constant used in the receiver's noise-temperature calculation
B. The EIRP radiated by the transmitting ground station or satellite
C. The total number of active transponders carried onboard the satellite
D. Free-space path loss together with rain/atmospheric attenuation ✓

Answer: D. Free-space path loss together with rain/atmospheric attenuation

94. In 5G NR, Frequency Range 1 (FR1) spans approximately:

A. 410 MHz to 7.125 GHz, the FR1 sub-6 GHz range ✓
B. 24 to 52 GHz, part of the FR2 mmWave range
C. 50 to 66 GHz, part of the FR2 upper mmWave extension
D. Below 100 MHz, in the HF/VHF broadcast range

Answer: A. 410 MHz to 7.125 GHz, the FR1 sub-6 GHz range

95. In 5G NR, a resource block (RB) in the frequency domain is defined as:

A. 14 consecutive subcarriers
B. 10 consecutive subframes
C. A single OFDM symbol
D. 12 consecutive subcarriers ✓

Answer: D. 12 consecutive subcarriers

96. A 5G NR radio frame lasts 10 ms and contains:

A. One subframe of 10 ms
B. Ten subframes of 1 ms each ✓
C. Fourteen subframes of 1 ms each
D. Twelve subframes of 0.5 ms each

Answer: B. Ten subframes of 1 ms each

97. In 5G NR, the subcarrier spacing given by delta-f = 2^u x 15 kHz means that it:

A. Remains fixed at 15 kHz for a typical sub-6 GHz 5G NR deployment
B. Decreases steadily as the numerology index u increases in value
C. Scales flexibly with the numerology u (e.g., 15, 30, 60, 120 kHz) ✓
D. Depends primarily on the carrier frequency band rather than the numerology

Answer: C. Scales flexibly with the numerology u (e.g., 15, 30, 60, 120 kHz)

98. Which set lists three 5G NR downlink physical channels?

A. PUSCH, PRACH and PUCCH
B. AMF, SMF and UPF
C. PDSCH, PDCCH and PBCH ✓
D. RRC, PDCP and RLC

Answer: C. PDSCH, PDCCH and PBCH

99. In the 5G initial access procedure, the UE first performs cell search using:

A. The random-access preamble on PRACH
B. The RRC Connection Setup message
C. The synchronization signals PSS and SSS ✓
D. The Sounding Reference Signal (SRS)

Answer: C. The synchronization signals PSS and SSS

100. To increase spectral efficiency, the 5G NR physical layer adopts which channel-coding techniques?

A. Convolutional coding and Reed-Solomon coding, inherited unchanged from earlier standards
B. Turbo coding, carried over unchanged from the LTE physical layer
C. No dedicated channel coding scheme, relying purely on modulation for robustness
D. Polar coding for control channels and LDPC coding for data channels ✓

Answer: D. Polar coding for control channels and LDPC coding for data channels

---

Part 2 - SatCom Payload, Links & 5G Systems

1. Which frequency band is most commonly used for mobile satellite phone handsets because it penetrates foliage well and suffers relatively little atmospheric attenuation?

A. Ka-band (26.5–40 GHz), which suffers severe rain fade and requires precise antenna pointing
B. V-band (40–75 GHz), which is heavily absorbed by atmospheric oxygen over long paths
C. mmWave FR2 (24–52 GHz), shared with terrestrial 5G small-cell deployments
D. L-band (approximately 1–2 GHz), which penetrates foliage and has low atmospheric attenuation ✓

Answer: D. L-band (approximately 1–2 GHz), which penetrates foliage and has low atmospheric attenuation

2. NTN is particularly well-suited for narrowband IoT (NB-IoT) deployments primarily because:

A. IoT devices typically require very high sustained bandwidth and continuous connectivity that broadband trunk satellite links, not NB-IoT waveforms, are built to provide
B. NTN reuses a dedicated satellite IoT protocol stack that was developed independently of 3GPP NB-IoT and cannot interoperate with terrestrial networks
C. IoT devices are often in remote areas without terrestrial coverage, and their low data-rate requirements are compatible with satellite link constraints ✓

D. IoT device battery capacity is too limited to survive the long acquisition and synchronisation time needed to access a GEO satellite link

Answer: C. IoT devices are often in remote areas without terrestrial coverage, and their low data-rate requirements are compatible with satellite link constraints

3. The Starlink constellation by SpaceX provides broadband primarily from which orbital shell?

A. GEO at 35,786 km, using a handful of large geostationary satellites over the equator

B. MEO at 20,200 km, mirroring the GPS navigation satellite constellation altitude

C. LEO at around 550 km, using thousands of small satellites for global broadband coverage ✓

D. HAPS at roughly 20 km altitude, using stratospheric balloons or solar aircraft

Answer: C. LEO at around 550 km, using thousands of small satellites for global broadband coverage

4. The Iridium system is notable for using inter-satellite links (ISLs) and for being able to route calls between satellites without a ground gateway underneath every satellite. Its constellation consists of:

A. 3 GEO satellites positioned over the equator, leaving the polar regions without coverage

B. 24 MEO satellites at GPS-like altitude, relying on a ground gateway for each individual call

C. Thousands of HAPS platforms in the stratosphere, each covering a small local area

D. 66 LEO satellites in polar orbit, providing near-global coverage including the poles ✓

Answer: D. 66 LEO satellites in polar orbit, providing near-global coverage including the poles

5. 3GPP Release 17 is significant to NTN because it:

A. Defined the baseline 5G NR air interface itself, which did not yet include any satellite-specific provisions

B. Introduced the first standardised 5G NR enhancements specifically for satellite access, including timing and Doppler adaptations ✓

C. Was the release that shelved the NTN study item proposed earlier, postponing satellite work indefinitely to a much later, unscheduled 3GPP release

D. Focused mainly on mmWave terrestrial small-cell enhancements, with satellite access left for a future release

Answer: B. Introduced the first standardised 5G NR enhancements specifically for satellite access, including timing and Doppler adaptations

6. Direct-to-Device (D2X) satellite services, which allow an unmodified smartphone to communicate via satellite, are enabled primarily by:

A. Proprietary satellite handsets sold directly by satellite operators, built years before 3GPP NTN specifications existed

B. Special hardware modules fitted at the factory that operate outside the scope of the 3GPP standard

C. New 3GPP NTN specifications that extend 5G NR for low-power handheld UEs accessing satellites directly ✓

D. A terrestrial base station relocated near the satellite ground track to relay signals to unmodified handsets

Answer: C. New 3GPP NTN specifications that extend 5G NR for low-power handheld UEs accessing satellites directly

7. A key advantage of NTN as a complement to terrestrial networks is that during large public events or disasters, NTN can:

A. Take over as the primary network indefinitely once terrestrial capacity is exceeded, retiring the terrestrial cells

B. Offload or supplement congested terrestrial cells, providing capacity relief and backup connectivity ✓

C. Deliver lower round-trip latency than a terrestrial macro cell, thanks to its wide beam footprint

D. Operate on the same spectrum as the terrestrial network without any coordination or interference management

Answer: B. Offload or supplement congested terrestrial cells, providing capacity relief and backup connectivity

8. As satellite altitude increases, the coverage footprint of a single satellite:

A. Increases, but at the cost of longer propagation delay and higher free-space path loss ✓

B. Decreases, because higher satellites must lower transmit power to stay within regulatory EIRP limits

C. Stays roughly the same across most altitudes, since footprint size is assumed to be set by the antenna beamwidth alone

D. Increases in proportion to altitude, while propagation delay stays fixed because delay is assumed to depend on frequency alone

Answer: A. Increases, but at the cost of longer propagation delay and higher free-space path loss

9. In the 6G vision, NTN is regarded as:

A. A stop-gap technology expected to be retired once dense urban 6G small-cell deployment is complete

B. A backhaul transport layer reserved for connecting gNBs, with no role in direct user access

C. A legacy 4G capability being phased out as networks migrate fully to 5G and 6G

D. A native component providing the 'coverage everywhere' pillar alongside terrestrial networks ✓

Answer: D. A native component providing the 'coverage everywhere' pillar alongside terrestrial networks

10. In a LEO NTN system, a UE may experience two types of handover: satellite-to-satellite (inter-satellite) and beam-to-beam within the same satellite (inter-beam). Both require:

A. Minimal signalling, since onboard satellite processing tracks each UE's position without network involvement

B. Mobility management procedures adapted to the rapidly changing satellite geometry ✓

C. Procedures borrowed directly from GEO systems, since LEO and GEO share an identical mobility standard

D. A physical reconnection of the UE's antenna cable to a different ground gateway station

Answer: B. Mobility management procedures adapted to the rapidly changing satellite geometry

11. Compared with GEO, LEO satellites offer lower latency because:

A. LEO satellites transmit at substantially higher power than GEO satellites, shortening the effective signal travel time

B. LEO satellites operate at a higher carrier frequency than GEO, and higher frequencies inherently propagate faster

C. Their altitude is roughly 550–1,500 km vs 35,786 km for GEO, greatly reducing one-way propagation delay ✓

D. LEO satellites are physically smaller and lighter than GEO satellites, which lets their signals travel faster through space

Answer: C. Their altitude is roughly 550–1,500 km vs 35,786 km for GEO, greatly reducing one-way propagation delay

12. Satellite link power limitation arises primarily because:

A. International spectrum regulations cap satellite EIRP at fixed limits regardless of the satellite's solar array size

B. The ground terminal, not the satellite payload, sets the transmit power level via remote command

C. The satellite's transmit power is constrained by the solar panel area and mass budget available in orbit ✓

D. Increasing transmit power tends to degrade signal quality because it drives the amplifier into saturation

Answer: C. The satellite's transmit power is constrained by the solar panel area and mass budget available in orbit

13. The large round-trip time of a GEO satellite link causes problems for the TCP protocol because:

A. TCP's slow-start mechanism ignores propagation delay by design, so GEO's RTT has no measurable effect on throughput

B. A large RTT causes the TCP congestion window to grow faster, which increases throughput on long-delay GEO links

C. The TCP congestion window fills slowly, capping throughput unless a Performance-Enhancing Proxy (PEP) or TCP optimisation is used ✓

D. TCP's original design assumed satellite-scale delays, so it performs noticeably better over GEO than over short terrestrial paths

Answer: C. The TCP congestion window fills slowly, capping throughput unless a Performance-Enhancing Proxy (PEP) or TCP optimisation is used

14. Spot beams in a satellite system concentrate transmit power onto a smaller geographic area in order to:

A. Lower the satellite's overall power consumption by shutting down amplifiers between transmissions

B. Increase the power spectral density within each beam and enable frequency reuse across spatially separated beams ✓

C. Illuminate the satellite's entire visible Earth disc simultaneously from one wide antenna element

D. Remove any need for frequency coordination between beams, since adjacent spot beams are assumed not to share the same frequency

Answer: B. Increase the power spectral density within each beam and enable frequency reuse across spatially separated beams

15. 3GPP defines two categories of NTN UE based on their ability to assist Doppler and timing compensation. These are:

A. Voice-oriented UEs and data-oriented UEs, categorised by the type of traffic they primarily carry

B. GNSS-capable UEs (which pre-compensate using their own position and ephemeris) and non-GNSS UEs (which rely on network assistance) ✓

C. Fixed UEs mounted at a stationary location and mobile UEs that move during a session

D. Transparent UEs that simply relay bent-pipe signals without processing and regenerative UEs that perform onboard demodulation and remodulation

Answer: B. GNSS-capable UEs (which pre-compensate using their own position and ephemeris) and non-GNSS UEs (which rely on network assistance)

16. Satellite spectrum coordination at the international level is managed primarily through:

A. WTO Basic Telecommunications Agreement tariff schedules, which govern market access rather than spectrum use

B. UN Security Council resolutions, which address international security matters rather than spectrum allocation

C. The 3GPP technical specification process, which standardises air-interface protocols rather than allocating spectrum

D. ITU Radio Regulations, bilateral and multilateral agreements between administrations ✓

Answer: D. ITU Radio Regulations, bilateral and multilateral agreements between administrations

17. In a bent-pipe (transparent) transponder, noise accumulated on the uplink is:

A. Amplified and retransmitted alongside the signal, degrading the overall downlink carrier-to-noise ratio ✓

B. Filtered out to a negligible level by the low-noise amplifier before the signal is retransmitted

C. Present mainly on the downlink hop, since the uplink path is assumed to be effectively noise-free

D. Removed by the frequency-conversion stage, which is assumed to strip noise along with shifting the carrier

Answer: A. Amplified and retransmitted alongside the signal, degrading the overall downlink carrier-to-noise ratio

18. Ephemeris data used by NTN UEs describes:

A. The satellite's onboard power budget, tracking how much energy remains in the batteries and solar arrays

B. The list of operators licensed to access the satellite's transponders under its regulatory authorisation, renewed on a periodic basis

C. The predicted position and velocity of the satellite over time, enabling the UE to compute Doppler shift and propagation delay ✓

D. The current software and firmware version running on the satellite's onboard payload processor

Answer: C. The predicted position and velocity of the satellite over time, enabling the UE to compute

Doppler shift and propagation delay

19. GNSS assists NTN UEs primarily by providing:

A. An alternative service link that substitutes for the satellite connection when visibility to the NTN satellite is poor

B. Accurate UE position, which combined with satellite ephemeris allows the UE to pre-compensate Doppler shift and timing advance ✓

C. A supply of electrical power to the UE's radio front-end, similar to wireless charging

D. A direct data path into the 5G core network, bypassing the NTN satellite and gNB radio path altogether

Answer: B. Accurate UE position, which combined with satellite ephemeris allows the UE to pre-compensate Doppler shift and timing advance

20. Maritime vessels in open ocean beyond terrestrial coastal coverage rely on NTN for:

A. Broadband internet, voice, safety and vessel-tracking services ✓

B. VHF terrestrial radio connections to land-based towers, which work solely within sight of the coastline

C. Fibre-optic undersea cable connections spliced directly into the vessel's onboard equipment

D. Microwave line-of-sight radio links relayed ship-to-ship until a coastal station is reached

Answer: A. Broadband internet, voice, safety and vessel-tracking services

21. In-flight connectivity (IFC) for commercial aircraft is provided by NTN because:

A. Aviation regulations require commercial aircraft to rely solely on terrestrial cellular towers for connectivity

B. Aircraft at cruise altitude are beyond the range of terrestrial networks, requiring satellite links for passenger and cockpit communications ✓

C. Aircraft fuselage-mounted antennas are physically incompatible with the frequency bands used by NTN satellites, requiring a redesigned antenna array

D. In-flight connectivity is delivered over long-range HF radio links rather than through satellite transponders

Answer: B. Aircraft at cruise altitude are beyond the range of terrestrial networks, requiring satellite links for passenger and cockpit communications

22. Circular polarization is preferred over linear polarization for mobile satellite terminals because:

A. Circular polarization experiences measurably less rain attenuation than linear polarization at the same frequency

B. The received signal polarization is independent of the terminal's physical orientation, simplifying the antenna design ✓

C. Linear polarization signals cannot be picked up by the feed horns used in typical satellite antennas

D. Circular polarization antennas achieve inherently higher peak gain than equivalent linear polarization antennas

Answer: B. The received signal polarization is independent of the terminal's physical orientation, simplifying the antenna design

23. The OneWeb LEO constellation differs from Starlink in that OneWeb focuses on:

A. Launching a fleet of GEO broadcast satellites aimed directly at consumer television subscribers

B. Operating a Ka-band constellation reserved for military and government customers under contracts that exclude any commercial retail service

C. Relying purely on bent-pipe payloads with no onboard IP routing, unlike Starlink's processed payloads

D. Providing broadband connectivity through a wholesale/B2B model, selling capacity to telecom operators rather than directly to consumers ✓

Answer: D. Providing broadband connectivity through a wholesale/B2B model, selling capacity to telecom operators rather than directly to consumers

24. A Performance-Enhancing Proxy (PEP) on a satellite gateway improves TCP throughput by:

A. Boosting the satellite's transmit power whenever congestion is detected, so packets arrive before timers expire
B. Replacing standard TCP with a proprietary transport protocol running solely on the UE, leaving the server unmodified
C. Splitting the TCP connection into a terrestrial segment and a satellite segment, hiding the large satellite RTT from the end-to-end TCP feedback loop ✓
D. Cutting the number of acknowledgement packets the server generates, though this does nothing to shorten the underlying satellite propagation delay

Answer: C. Splitting the TCP connection into a terrestrial segment and a satellite segment, hiding the large satellite RTT from the end-to-end TCP feedback loop

25. Which NTN platform offers the most rapid deployment and easiest recovery after failure, but is most limited by weather and flight regulations?

A. GEO satellite, which ground controllers can reposition to a new orbital slot within a few hours
B. LEO satellite, which operators can temporarily lower to an altitude of about 50 km on demand
C. Unmanned Aircraft System (UAS/drone) used as an airborne NTN node ✓
D. HAPS balloon, which stays fully stable and unaffected by strong upper-atmosphere winds

Answer: C. Unmanned Aircraft System (UAS/drone) used as an airborne NTN node

26. Free-space path loss (FSPL) increases with distance and frequency according to FSPL(dB) = 20 log(d) + 20 log(f) + constant. A doubling of the carrier frequency therefore increases FSPL by approximately:

A. 3 dB
B. 10 dB
C. 6 dB ✓
D. 0 dB — frequency has no effect on free-space path loss

Answer: C. 6 dB

27. Rain attenuation is most severe at which frequency range?

A. Below 2 GHz (L-band), where the wavelength is much larger than a raindrop and scattering is negligible
B. At optical and infrared wavelengths, where clouds absorb light heavily but microwave links pass through unaffected
C. Above approximately 10 GHz (Ku, Ka and higher bands), where raindrops are comparable in size to the signal wavelength ✓
D. Between 400 and 700 MHz (UHF), a band where atmospheric water vapour absorption happens to peak

Answer: C. Above approximately 10 GHz (Ku, Ka and higher bands), where raindrops are comparable in size to the signal wavelength

28. A satellite orbit's inclination is defined as:

A. The satellite's altitude above mean sea level, measured at the point directly below the spacecraft
B. The angle between the orbital plane and the Earth's equatorial plane; 0° is equatorial (GEO), 90° is polar ✓
C. The eccentricity of the orbital ellipse, describing how stretched the orbit is compared with a perfect circle
D. The angle between the satellite's antenna boresight and the local Earth surface tangent

Answer: B. The angle between the orbital plane and the Earth's equatorial plane; 0° is equatorial (GEO), 90° is polar

29. A sun-synchronous orbit is used by Earth-observation satellites because:

A. The satellite actually orbits the Sun directly rather than the Earth, giving it continuous solar exposure
B. The satellite remains fixed above the same point on the Earth's surface continuously, like a geostationary satellite in an equatorial orbit
C. The satellite's solar panels are mechanically fixed perpendicular to the Sun for the entire mission by orbit design alone
D. The orbital plane precesses at the same rate as the Earth's revolution around the Sun,

keeping the local solar illumination time constant over each pass ✓

Answer: D. The orbital plane precesses at the same rate as the Earth's revolution around the Sun, keeping the local solar illumination time constant over each pass

30. The Van Allen radiation belts are a concern for satellite designers because:

A. They are zones of trapped high-energy charged particles that can degrade semiconductor electronics and solar cells ✓

B. They are regions of charged dust that cause rain-like attenuation on Ku-band downlink signals

C. They form a shielding layer that blocks most radio transmissions from reaching geostationary altitude

D. They are dense cloud layers in the upper troposphere that scatter and absorb microwave signals

Answer: A. They are zones of trapped high-energy charged particles that can degrade semiconductor electronics and solar cells

31. The ionosphere (roughly 60–1,000 km altitude) affects satellite radio signals primarily by causing:

A. Rain fade and tropospheric scintillation, effects that actually occur above about 10 GHz in the lower troposphere

B. Inter-symbol interference caused by multipath reflections off buildings near the ground terminal

C. Propagation delay, Faraday rotation of polarization, and signal dispersion, particularly at frequencies below about 3 GHz ✓

D. An increase in thermal noise generated internally within the satellite's low-noise amplifier stage

Answer: C. Propagation delay, Faraday rotation of polarization, and signal dispersion, particularly at frequencies below about 3 GHz

32. Tropospheric scintillation is a rapid fluctuation of received signal amplitude and phase caused by turbulence in the lower atmosphere. It is most significant at:

A. High elevation angles and low frequencies below 1 GHz, where the atmospheric path is shortest

B. A broad range of elevation angles and frequencies about equally, since turbulence is assumed uniform throughout the troposphere

C. Low elevation angles and frequencies above about 10 GHz ✓

D. Polar latitudes specifically, where the troposphere is thinner and turbulence is assumed weaker

Answer: C. Low elevation angles and frequencies above about 10 GHz

33. The Doppler frequency shift experienced by a ground receiver tracking a LEO satellite depends primarily on:

A. The radial component of the satellite's velocity relative to the receiver, and the carrier frequency ✓

B. The satellite's altitude alone, since Doppler shift is assumed to depend on height rather than relative velocity

C. The carrier frequency alone, treating the satellite's motion as having no bearing on the observed shift

D. The signal's polarization state rather than the relative geometry between satellite and ground receiver

Answer: A. The radial component of the satellite's velocity relative to the receiver, and the carrier frequency

34. A GEO satellite produces negligible Doppler shift for a stationary ground terminal because:

A. GEO satellites orbit slowly enough that their absolute velocity, not their relative velocity to the ground, is assumed to be effectively zero in a typical mission scenario

B. Both the satellite and the terminal are effectively stationary relative to each other — the satellite's angular velocity matches the Earth's rotation ✓

C. Doppler shift is actively cancelled inside the transponder's frequency-conversion stage before retransmission

D. GEO satellites broadcast a dedicated pilot tone that ground receivers use to null out any residual Doppler shift

Answer: B. Both the satellite and the terminal are effectively stationary relative to each other — the

satellite's angular velocity matches the Earth's rotation

35. The elevation angle of a satellite as seen from a ground station determines:

A. The satellite's orbital altitude above mean sea level, independent of the ground station's location

B. The satellite's velocity relative to the ground station, which sets the Doppler shift rather than the path length

C. The total number of other satellites simultaneously visible in the sky from that ground station

D. The length of the signal path through the atmosphere; higher elevation means a shorter atmospheric path and less attenuation ✓

Answer: D. The length of the signal path through the atmosphere; higher elevation means a shorter atmospheric path and less attenuation

36. Link margin in a satellite system is defined as:

A. The physical diameter of the satellite's main reflector antenna, which sets gain but is not itself a margin figure

B. The maximum data rate the link is permitted to carry under the operator's spectrum licence

C. The ratio of transmit power to receive power measured at the ground station's demodulator input

D. The excess of received Eb/N0 (or C/N0) above the minimum threshold required for the specified bit-error rate ✓

Answer: D. The excess of received Eb/N0 (or C/N0) above the minimum threshold required for the specified bit-error rate

37. During equinox periods, GEO satellites experience solar eclipses lasting up to about:

A. 72 minutes per day, during which the satellite relies on onboard batteries ✓

B. 12 hours per day, on the assumption that the satellite sits behind the Earth for the entire local night

C. Under 1 minute per day, treating the Earth's shadow at GEO altitude as extremely narrow

D. GEO satellites are assumed to avoid eclipse because their equatorial orbital plane is thought to keep them continuously in sunlight

Answer: A. 72 minutes per day, during which the satellite relies on onboard batteries

38. Frequency reuse in a multi-beam satellite system increases overall system capacity because:

A. Each beam is assigned a distinct frequency, so reuse merely describes the repeating pattern of the spectrum plan

B. Frequency reuse is assumed to reduce co-channel interference to a negligible, effectively zero level

C. The same frequency band is reused across spatially separated beams, multiplying the total traffic the satellite can carry ✓

D. Total system capacity is fixed by the number of beams alone, irrespective of how frequencies are reused among them

Answer: C. The same frequency band is reused across spatially separated beams, multiplying the total traffic the satellite can carry

39. When the same frequency is used in adjacent satellite beams (co-channel reuse), co-channel interference (CCI) is controlled by:

A. Sufficient beam spacing and antenna sidelobe suppression to keep the interfering signal below acceptable levels ✓

B. Raising the carrier frequency progressively until the coverage of adjacent beams no longer geometrically overlaps

C. Assigning a distinct polarization to each individual beam across the whole satellite footprint

D. Forcing each pair of adjacent beams to transmit at precisely matched power levels regardless of traffic load

Answer: A. Sufficient beam spacing and antenna sidelobe suppression to keep the interfering signal below acceptable levels

40. By Kepler's third law, the orbital period of a satellite is proportional to the cube root of the cube of the semi-major axis ($T^2 \propto a^3$). This means:

A. Higher orbits have shorter orbital periods and correspondingly higher orbital speeds than lower orbits

B. The orbital period stays roughly constant across most altitudes, since Kepler's law is assumed to depend on eccentricity rather than altitude

C. Higher orbits have longer periods and lower orbital speeds than lower orbits ✓

D. The orbital period depends primarily on the satellite's own mass rather than on its distance from Earth

Answer: C. Higher orbits have longer periods and lower orbital speeds than lower orbits

41. GEO satellites have poor coverage poleward of approximately ±75° latitude because:

A. The Van Allen radiation belts are assumed to block radio propagation specifically over the polar regions

B. International spectrum regulations are assumed to prohibit GEO satellite transmissions above 75° latitude

C. They orbit in the equatorial plane, so the elevation angle seen from high latitudes is very low, below the horizon or near it ✓

D. Ionospheric absorption near the poles is assumed to be so severe that essentially no signal reaches the ground

Answer: C. They orbit in the equatorial plane, so the elevation angle seen from high latitudes is very low, below the horizon or near it

42. A Molniya orbit is a highly elliptical orbit (HEO) with an inclination of about 63.4° and a 12-hour period. Its primary purpose is to:

A. Provide a geostationary-like fixed view of the equatorial belt, much as a standard GEO satellite would

B. Achieve the shortest possible propagation delay for users near the Arctic Circle, shorter than any LEO system

C. Provide long-dwell coverage of high-latitude regions (such as Russia) where GEO elevation angles are too low ✓

D. Allow a satellite to cross the equatorial plane several times each day while remaining at a fixed inclination

Answer: C. Provide long-dwell coverage of high-latitude regions (such as Russia) where GEO elevation angles are too low

43. The Iridium constellation routes calls via inter-satellite links (ISLs), which means:

A. Calls between any two points on Earth can be routed entirely through the satellite network without requiring a ground gateway in between ✓

B. Each call must still be relayed through a dedicated ground gateway at each satellite-to-satellite hop it passes through

C. Iridium's inter-satellite links are reserved for housekeeping telemetry, and subscriber voice traffic must instead route through gateways

D. Iridium's inter-satellite links operate in the optical band, a medium not capable of carrying real-time voice traffic

Answer: A. Calls between any two points on Earth can be routed entirely through the satellite network without requiring a ground gateway in between

44. The ITU-R rain zone classification system (zones A to Q) is used to:

A. Set the maximum allowable satellite transmit power permitted within each geographic rain zone

B. Classify which geostationary orbital slots are available for satellites registered under each rain zone administration

C. Define which frequency bands are allocated to different satellite services within each rain zone

D. Estimate the rain attenuation statistics for a given geographic location and design satellite link margins accordingly ✓

Answer: D. Estimate the rain attenuation statistics for a given geographic location and design satellite link margins accordingly

45. A multi-beam satellite antenna produces higher gain per beam than a global beam antenna. The trade-off is that:

A. Each individual beam consumes disproportionately more onboard power, which actually reduces the satellite's total capacity

B. Multi-beam antennas are structurally unable to reuse the same frequency band across different beams

C. Achieving higher gain per beam inherently lowers the overall throughput the whole satellite system can deliver

D. Each beam covers a smaller geographic area, requiring more beams to achieve the same total coverage ✓

Answer: D. Each beam covers a smaller geographic area, requiring more beams to achieve the same total coverage

46. The geosynchronous orbit is the set of all orbits with a 24-hour period, while the geostationary orbit (GEO) is a subset defined by:

A. Zero inclination (equatorial plane) and zero eccentricity (circular orbit), resulting in a fixed position above the equator ✓

B. A highly elliptical orbit inclined at 63.4°, the same geometry used by Molniya-type satellites

C. Any polar orbit that happens to complete one revolution each 24-hour period, regardless of eccentricity

D. An orbital inclination of exactly 28.5°, matching the latitude of major equatorial launch sites such as Cape Canaveral

Answer: A. Zero inclination (equatorial plane) and zero eccentricity (circular orbit), resulting in a fixed position above the equator

47. Satellite eclipse occurs when the Earth's shadow blocks sunlight from reaching the satellite. During eclipse, the satellite continues operating because:

A. Solar panels are assumed to store energy chemically within themselves, removing any need for separate batteries

B. A backup radioisotope power unit is assumed to switch on once eclipse begins, without needing stored battery charge

C. The satellite shuts its payload power down to zero during eclipse and simply waits for sunlight to return

D. Onboard batteries are charged during sunlit periods and supply power during eclipse ✓

Answer: D. Onboard batteries are charged during sunlit periods and supply power during eclipse

48. The figure of merit G/T of a satellite receiver system improves (increases) when:

A. Antenna receive gain G increases or system noise temperature T decreases ✓

B. Path loss along the link increases, which is unrelated to the receiver's own gain-to-noise-temperature figure

C. The transmitting station's EIRP decreases, a transmit-side parameter that does not enter the G/T figure

D. The carrier frequency is lowered while the antenna gain G and noise temperature T are both held fixed

Answer: A. Antenna receive gain G increases or system noise temperature T decreases

49. In the link budget of a satellite system, the overall end-to-end performance is determined by:

A. The direct sum of the uplink and downlink C/N ratios once both are converted to linear scale

B. The uplink alone, on the assumption that the satellite's transponder amplifies the downlink without adding any noise

C. The weaker of the uplink and downlink, since the total signal-to-noise ratio is dominated by the noisier hop ✓

D. The downlink alone, on the assumption that the uplink carries a comfortable margin in most practical link designs

Answer: C. The weaker of the uplink and downlink, since the total signal-to-noise ratio is dominated by the noisier hop

50. Polarization diversity allows a satellite to reuse the same frequency band twice by transmitting two independent signals using:

A. Two separate carrier frequencies transmitted using the same single polarization

B. Orthogonal polarizations (e.g., horizontal and vertical linear, or RHCP and LHCP circular) ✓

C. Two copies of the identical signal transmitted with a 90-degree phase offset rather than orthogonal polarization

D. Two separate antennas physically pointed at two different geographic coverage areas

Answer: B. Orthogonal polarizations (e.g., horizontal and vertical linear, or RHCP and LHCP circular)

51. 3GPP Release 17 introduced the first NTN-specific 5G NR enhancements, which included adaptations to timing advance, Doppler compensation and:

A. Removal of the standard OFDM waveform, replaced by a bespoke satellite-specific modulation scheme

B. Allocation of a brand-new dedicated NTN frequency band located below 100 MHz

C. Replacement of the standard 5G core network with a separate satellite-specific core architecture

D. HARQ process design to handle the long satellite round-trip time ✓

Answer: D. HARQ process design to handle the long satellite round-trip time

52. In NTN, an extended Timing Advance (TA) mechanism is required because:

A. The satellite's onboard clock is assumed to run measurably faster than the ground station's reference clock

B. Timing advance is applied to delay the downlink signal so that it matches the user's ground location

C. Propagation delays of tens to hundreds of milliseconds must be pre-compensated so that uplink transmissions arrive at the gNB within the correct slot boundary ✓

D. Timing advance is needed solely for stationary UEs and is switched off the moment a UE begins moving, unlike the fixed procedure used in terrestrial NR

Answer: C. Propagation delays of tens to hundreds of milliseconds must be pre-compensated so that uplink transmissions arrive at the gNB within the correct slot boundary

53. The PRACH (Physical Random Access Channel) preamble design is modified for NTN by:

A. Extending the preamble length and guard period to accommodate the large round-trip propagation delay, preventing preamble collisions ✓

B. Shortening the preamble and guard period to cut overhead, on the assumption that a shorter window suits long delays

C. Replacing the Zadoff-Chu preamble sequence with a CDMA spreading code borrowed from 3G systems

D. Moving the PRACH transmission to the downlink direction, where the gNB rather than the UE originates it

Answer: A. Extending the preamble length and guard period to accommodate the large round-trip propagation delay, preventing preamble collisions

54. The fundamental problem HARQ faces in a GEO NTN system is that:

A. HARQ acknowledgements are individually encrypted, and the added decryption step is what causes the delay

B. HARQ retransmissions are assumed to travel back to the UE faster than the original transmission did

C. The round-trip time over a GEO link is assumed to match the roughly 8-ms terrestrial NR RTT, leaving HARQ buffering requirements unaffected

D. The 500+ ms RTT means the transmitter must buffer many HARQ processes simultaneously, far exceeding the 8-ms terrestrial HARQ round-trip budget ✓

Answer: D. The 500+ ms RTT means the transmitter must buffer many HARQ processes simultaneously, far exceeding the 8-ms terrestrial HARQ round-trip budget

55. For GEO NTN, 3GPP permits HARQ to be disabled. When HARQ feedback is turned off, residual errors are handled by:

A. The satellite's onboard processor, which is assumed to correct bit errors itself before relaying the signal

B. Outer-layer error recovery mechanisms such as RLC ARQ or application-layer FEC ✓

C. Progressively increasing the UE's transmit power until the error rate is driven down to nothing

D. A fixed rule that repeats each transmission exactly twice, independent of whether an error actually occurred

Answer: B. Outer-layer error recovery mechanisms such as RLC ARQ or application-layer FEC

56. As a LEO satellite moves, beam management in NTN must handle:

A. Static beam assignments that stay fixed for the duration of a session, regardless of satellite motion

B. Uplink beam management alone, on the assumption that the downlink beam remains optimal without adjustment

C. Beam management procedures carried over unchanged from terrestrial massive-MIMO base stations, which assume a fixed antenna array location

D. Beam sweeping, beam measurement reporting, and beam failure recovery adapted to the satellite's high angular velocity relative to the ground ✓

Answer: D. Beam sweeping, beam measurement reporting, and beam failure recovery adapted to the satellite's high angular velocity relative to the ground

57. In a LEO NTN with Earth-moving beams, the cell associated with a given geographic area changes as the satellite passes overhead. 3GPP addresses this by:

A. Distinguishing between Earth-fixed and Earth-moving cell identities, and adapting cell re-selection and handover procedures accordingly ✓

B. Assigning a fixed, permanent cell identity to each geographic point on Earth, independent of which satellite passes overhead

C. Requiring the UE to fall back temporarily to a terrestrial network during each cell transition

D. Removing the cell concept from NTN altogether and replacing it with continuous satellite-based tracking

Answer: A. Distinguishing between Earth-fixed and Earth-moving cell identities, and adapting cell re-selection and handover procedures accordingly

58. Conditional Handover (CHO) is particularly beneficial in NTN because:

A. The network can prepare the target cell in advance and trigger execution automatically when conditions are met, avoiding added signalling latency on a long-RTT link ✓

B. Conditional handover is assumed to require a very short RTT, which would actually suit GEO rather than LEO

C. Conditional handover is prohibited under NTN specifications, which instead rely on fast re-registration procedures

D. Conditional handover applies solely to UEs that are physically moving on the ground, and offers no benefit against satellite-induced signalling latency

Answer: A. The network can prepare the target cell in advance and trigger execution automatically when conditions are met, avoiding added signalling latency on a long-RTT link

59. The primary frequency bands targeted by 3GPP NR-NTN specifications for the service link (UE-to-satellite) are:

A. S-band (around 2 GHz) and Ka-band (around 26 GHz) ✓

B. mmWave FR2 bands (24–52 GHz) shared directly with terrestrial 5G small cells

C. HF bands (3–30 MHz), chosen for their long-range ionospheric propagation characteristics

D. The same Ku-band frequencies used for the satellite's feeder link to the ground gateway

Answer: A. S-band (around 2 GHz) and Ka-band (around 26 GHz)

60. Inter-satellite links (ISLs) are most naturally associated with the regenerative payload architecture because:

A. Bent-pipe payloads are assumed to support inter-satellite links natively, since they need essentially no onboard signal processing

B. A regenerative satellite can route and forward data packets, making it practical to relay traffic between satellites toward a distant gateway ✓

C. Inter-satellite links are assumed to require the transparent frequency conversion stage that bent-pipe payloads are specifically built to provide

D. ITU regulations are assumed to prohibit inter-satellite links between satellites carrying regenerative payloads

Answer: B. A regenerative satellite can route and forward data packets, making it practical to relay traffic between satellites toward a distant gateway

61. In the RLC layer, the NTN large RTT creates an issue analogous to the TCP window problem, because:

A. The RLC ARQ window may need to be enlarged to allow enough outstanding unacknowledged PDUs so that throughput is not stalled waiting for acknowledgements ✓
B. RLC ARQ is switched off for NTN links, which removes any window-size constraint from the resulting throughput calculation
C. The RLC window size is assumed to have no bearing on achievable throughput, regardless of the link's RTT
D. The RLC sublayer itself was removed from the NTN protocol stack as part of the Release 17 specifications

Answer: A. The RLC ARQ window may need to be enlarged to allow enough outstanding unacknowledged PDUs so that throughput is not stalled waiting for acknowledgements

62. In LEO NTN, paging areas tend to be larger than in terrestrial networks because:

A. LEO satellites are assumed to have limited onboard resources, capping the number of UEs that can be paged at once
B. Satellite footprints cover vast geographic areas, and using large paging areas reduces the frequency of tracking area updates as the satellite moves ✓
C. Larger paging areas are assumed to reduce the total data volume that needs to be transmitted to each UE
D. Paging in NTN uses procedures and area sizes carried over unchanged from terrestrial 5G networks, ignoring the satellite footprint size

Answer: B. Satellite footprints cover vast geographic areas, and using large paging areas reduces the frequency of tracking area updates as the satellite moves

63. Open-loop power control is preferred over closed-loop power control for NTN uplinks because:

A. Closed-loop control is assumed to be more accurate but to demand extra dedicated hardware inside the UE
B. The long RTT makes it impractical to rely on fast gNB feedback to adjust UE transmit power in near real-time ✓
C. Open-loop control is assumed to rely on gNB feedback that arrives within about 1 millisecond, like terrestrial NR
D. 3GPP specifications are assumed to mandate closed-loop power control as the default for NTN uplink transmissions

Answer: B. The long RTT makes it impractical to rely on fast gNB feedback to adjust UE transmit power in near real-time

64. eMBB (enhanced Mobile BroadBand) services over LEO NTN are achievable, but their throughput is typically lower than terrestrial 5G because of:

A. LEO NTN relying on a fundamentally different air-interface waveform that is incompatible with standard eMBB service
B. 3GPP specifications explicitly prohibiting the delivery of eMBB services over NTN in Release 17
C. eMBB service being defined as requiring sub-millisecond latency, a target no NTN satellite link can realistically reach
D. Large free-space path loss and the constrained transmit power of the satellite, limiting link capacity ✓

Answer: D. Large free-space path loss and the constrained transmit power of the satellite, limiting link capacity

65. URLLC (Ultra-Reliable Low-Latency Communication) requirements cannot be met by GEO NTN primarily because:

A. GEO satellites lack sufficient transmit power to carry the data rates URLLC services typically demand
B. GEO round-trip delay (~600 ms) far exceeds the sub-millisecond latency target of URLLC ✓
C. URLLC is defined solely as a terrestrial 5G feature and is formally excluded from the NTN specifications
D. GEO NTN mandates the use of HARQ retransmission, a mechanism assumed to be fundamentally incompatible with URLLC

Answer: B. GEO round-trip delay (~600 ms) far exceeds the sub-millisecond latency target of URLLC

66. Larger subcarrier spacing (higher numerology) in 5G NR is more robust against Doppler-induced inter-carrier interference in NTN because:

A. Larger subcarrier spacing shortens the OFDM symbol duration, which is assumed to give the channel more time to change within a symbol

B. A wider subcarrier spacing means the Doppler shift is a smaller fraction of the subcarrier bandwidth, reducing ICI ✓

C. Lower numerology, meaning smaller subcarrier spacing, is assumed to be the generally preferred configuration for NTN links

D. Doppler shift is assumed to be unrelated to subcarrier spacing in any OFDM-based system

Answer: B. A wider subcarrier spacing means the Doppler shift is a smaller fraction of the subcarrier bandwidth, reducing ICI

67. Gateway diversity in NTN means:

A. Multiple geographically separated gateways can serve the same satellite, so if one suffers rain fade the satellite can switch to a clear-sky gateway automatically ✓

B. Deploying multiple satellites to serve a single ground gateway simultaneously, rather than diversifying the gateway sites themselves geographically

C. Installing redundant user terminals at each individual subscriber site to guard against local outages

D. Routing gateway traffic through two fully separate 5G core network instances for redundancy

Answer: A. Multiple geographically separated gateways can serve the same satellite, so if one suffers rain fade the satellite can switch to a clear-sky gateway automatically

68. Ka-band is attractive for high-throughput satellite (HTS) systems because:

A. Ka-band signals are assumed to be largely unaffected by rain, which would make the band especially well suited to tropical deployments

B. Wide available bandwidths (~500 MHz or more per polarization) enable high throughput, despite more severe rain attenuation and stricter antenna pointing requirements than L- or S-band ✓

C. Ka-band is assumed to offer narrower channel bandwidths than L-band, which would make it more spectrally efficient

D. Ka-band spectrum is assumed to be reserved solely for military satellite communications, excluding commercial use

Answer: B. Wide available bandwidths (~500 MHz or more per polarization) enable high throughput, despite more severe rain attenuation and stricter antenna pointing requirements than L- or S-band

69. NTN is well-suited for massive Machine-Type Communications (mMTC) IoT primarily because:

A. IoT sensors are assumed to demand ultra-high bandwidth that solely satellite trunk links are capable of supplying

B. mMTC is assumed to require sub-millisecond latency, a target that specifically low-altitude LEO satellites are said to reliably reach

C. IoT devices are assumed to be unable to connect to any satellite because they lack an onboard GNSS receiver

D. IoT sensors are often deployed in remote or unpowered locations without terrestrial coverage, and their low data-rate and infrequent transmission requirements suit satellite capacity constraints ✓

Answer: D. IoT sensors are often deployed in remote or unpowered locations without terrestrial coverage, and their low data-rate and infrequent transmission requirements suit satellite capacity constraints

70. When a UE moves between satellite beams or between satellites, 5G session continuity mechanisms ensure:

A. PDU sessions are fully terminated and re-established from scratch each time the UE changes satellite or beam

B. PDU sessions are maintained without application-level disruption, through UPF selection and session anchor management in the 5G core ✓

C. Session continuity is managed solely by the UE's own software, without any coordination from the 5G core network

D. Session continuity mechanisms are assumed to cover voice calls specifically, while data sessions are simply dropped

Answer: B. PDU sessions are maintained without application-level disruption, through UPF selection and session anchor management in the 5G core

71. A GNSS-capable NTN UE uses its own position and satellite ephemeris to:

A. Authenticate itself directly with the satellite's onboard processor before being allowed to transmit

B. Determine which 5G core network slice the UE should connect to for its intended service

C. Request a handover to a terrestrial network whenever a GNSS fix becomes available, regardless of the current NTN link quality or signal strength

D. Pre-compute the uplink frequency offset and timing advance needed to compensate for Doppler shift and propagation delay before transmitting ✓

Answer: D. Pre-compute the uplink frequency offset and timing advance needed to compensate for Doppler shift and propagation delay before transmitting

72. Timing pre-compensation in NTN requires the UE to:

A. Delay, rather than advance, its uplink transmission to compensate for the satellite's forward orbital motion

B. Advance its uplink transmission by an amount equal to the computed propagation delay to the satellite, so the signal arrives within the expected slot window ✓

C. Request that the gNB shift the downlink timing by the propagation delay instead of adjusting its own uplink

D. Transmit using a fixed timing offset of 0.5 ms that stays constant no matter how far away the satellite currently is in its orbit

Answer: B. Advance its uplink transmission by an amount equal to the computed propagation delay to the satellite, so the signal arrives within the expected slot window

73. Network slicing in 5G NTN allows different service types (broadband, IoT, emergency) to share the same NTN physical resources while maintaining:

A. Independent quality of service, isolation and scheduling policies per slice ✓

B. One shared QoS profile applied across each service type, chosen to simplify the network's overall management

C. A dedicated physical satellite payload allocated separately to each individual network slice

D. No meaningful service differentiation, with each slice receiving essentially identical scheduling treatment

Answer: A. Independent quality of service, isolation and scheduling policies per slice

74. 5G NR reference signals (DMRS, CSI-RS, SRS) in NTN must be designed to account for:

A. The large and time-varying delays of the NTN channel, requiring additional reference signal configurations for accurate channel estimation ✓

B. The satellite channel's assumed complete absence of multipath, unlike dense urban terrestrial deployments

C. The assumption that the NTN channel is perfectly static over time, meaning the standard terrestrial channel estimation approach applies unchanged

D. A shorter OFDM symbol duration than terrestrial NR uses, intended to fit in more pilot reference symbols

Answer: A. The large and time-varying delays of the NTN channel, requiring additional reference signal configurations for accurate channel estimation

75. The main factor limiting uplink throughput for direct-to-handset LEO NTN is:

A. The downlink direction, which is assumed to be the bottleneck across most NTN system designs regardless of UE type

B. Excessive UE transmit power that is assumed to spill over and cause interference with neighbouring satellite constellations sharing the band

C. The constrained transmit power of handheld UEs (typically 23–26 dBm), which combined with the large path loss sets an upper bound on achievable uplink data rate ✓

D. The lack of multi-antenna MIMO support at the handset, which is assumed to force it into single-antenna transmission mode

Answer: C. The constrained transmit power of handheld UEs (typically 23–26 dBm), which combined with the large path loss sets an upper bound on achievable uplink data rate

76. DVB-S2 (Digital Video Broadcasting — Satellite, Second Generation) is significant for satellite broadband because it:

A. Employs Adaptive Coding and Modulation (ACM), dynamically selecting the modulation and coding scheme to maximise throughput under varying link conditions ✓
B. Uses a fixed QPSK modulation scheme at a constant code rate, chosen deliberately to keep receiver design simple
C. Is a 3GPP-defined standard, developed specifically as part of the 5G NR satellite access specifications
D. Is fundamentally a terrestrial digital broadcast standard, unrelated in design to any of the DVB satellite broadcast specifications

Answer: A. Employs Adaptive Coding and Modulation (ACM), dynamically selecting the modulation and coding scheme to maximise throughput under varying link conditions

77. DVB-RCS2 (Return Channel via Satellite, Second Generation) defines:

A. The forward broadcast link running from the satellite outward to each subscriber terminal in the coverage footprint
B. The interactive return channel from the user terminal to the satellite, enabling two-way broadband satellite services ✓
C. The ITU standard governing how administrations coordinate geostationary orbital slot assignments
D. The inter-satellite link protocol adopted by LEO constellations such as Iridium and Starlink

Answer: B. The interactive return channel from the user terminal to the satellite, enabling two-way broadband satellite services

78. A VSAT (Very Small Aperture Terminal) system typically uses:

A. Small dish antennas of 0.6–3.6 m diameter to provide satellite broadband at fixed or mobile sites, usually via GEO satellites in Ku or Ka band ✓
B. Large parabolic dishes exceeding 10 metres in diameter, typically reserved for military ground stations and deep-space tracking facilities
C. LEO satellite constellations alone, since VSAT terminals are assumed unable to link to any GEO satellite
D. Phased-array antenna panels mounted on aircraft or balloons flying above 50 km altitude

Answer: A. Small dish antennas of 0.6–3.6 m diameter to provide satellite broadband at fixed or mobile sites, usually via GEO satellites in Ku or Ka band

79. In FDMA (Frequency Division Multiple Access) for satellite systems, a key disadvantage is:

A. Spectrum is wasted when an assigned frequency sub-band is idle because its allocated user has no traffic to send ✓
B. Each user is forced to transmit within the same shared time slot, which causes frequent collisions
C. FDMA is assumed to require tighter synchronization than any other multiple-access scheme, including TDMA
D. FDMA is assumed to be structurally incompatible with combining alongside TDMA on the same satellite transponder

Answer: A. Spectrum is wasted when an assigned frequency sub-band is idle because its allocated user has no traffic to send

80. Satellite TDMA (Time Division Multiple Access) requires tight time synchronization across all terminals because:

A. TDMA assigns a unique spreading code to each user, and synchronisation is what prevents those codes from colliding
B. TDMA gives each terminal a separate carrier frequency, and timing synchronisation is what prevents frequency drift
C. Without synchronisation, bursts from different terminals will overlap in time at the satellite, causing interference ✓
D. Synchronisation is required solely at the ground gateway, since individual user terminals need no separate timing reference of their own

Answer: C. Without synchronisation, bursts from different terminals will overlap in time at the satellite, causing interference

81. In CDMA (Code Division Multiple Access), all users share the same bandwidth simultaneously and are separated by unique spreading codes. A critical operational requirement is:

A. That no two users are ever scheduled to transmit within the same time slot, as in TDMA
B. That each user is instead assigned a dedicated separate frequency channel, as in FDMA
C. That the spreading codes assigned to each user are regenerated on a fixed one-millisecond cycle
D. Accurate uplink power control, so that no single user's signal overpowers the others (the near-far problem) ✓

Answer: D. Accurate uplink power control, so that no single user's signal overpowers the others (the near-far problem)

82. OFDMA (Orthogonal Frequency Division Multiple Access) is used in the 5G NR downlink because it:

A. Assigns each user a unique CDMA-style spreading code rather than a set of frequency subcarriers
B. Is essentially FDMA implemented with analogue carriers rather than digitally generated subcarriers
C. Was originally designed for single-user point-to-point links and lacks any native support for multi-user scheduling decisions
D. Allows flexible per-slot scheduling of subcarriers to different users, enabling efficient multi-user resource allocation ✓

Answer: D. Allows flexible per-slot scheduling of subcarriers to different users, enabling efficient multi-user resource allocation

83. DFT-spread OFDM (DFT-s-OFDM), also called SC-FDMA, is used in the 5G NR uplink rather than pure OFDMA because it:

A. Actually produces a higher peak-to-average power ratio than plain OFDM, which is assumed to suit power amplifiers better
B. Is mandated specifically for satellite gNB transmitters while being formally prohibited across terrestrial gNB uplink deployments worldwide
C. Spreads each user's data across fewer subcarriers than OFDMA, which is assumed to shrink the transmitted bandwidth
D. Achieves a lower Peak-to-Average Power Ratio (PAPR), which reduces power amplifier stress and is beneficial for power-constrained UE transmitters ✓

Answer: D. Achieves a lower Peak-to-Average Power Ratio (PAPR), which reduces power amplifier stress and is beneficial for power-constrained UE transmitters

84. The Access and Mobility Management Function (AMF) in the 5G SA core is responsible for:

A. Routing user-plane IP packets directly between the radio access network and the public internet
B. Establishing, modifying and releasing PDU sessions along with allocating IP addresses to UEs
C. UE registration, connection establishment and mobility management including handover decisions ✓
D. Scheduling individual radio resource blocks on the air interface inside the gNB scheduler

Answer: C. UE registration, connection establishment and mobility management including handover decisions

85. The Session Management Function (SMF) in the 5G SA core is responsible for:

A. Handling UE authentication and initial registration procedures with the 5G core network
B. Performing physical-layer radio resource scheduling decisions inside the gNB itself
C. Establishing, modifying and releasing PDU sessions and managing IP address allocation for UEs ✓
D. Providing the physical NG interface connection that links the gNB to the 5G core network

Answer: C. Establishing, modifying and releasing PDU sessions and managing IP address allocation for UEs

86. The User Plane Function (UPF) in the 5G SA core is the:

A. Control-plane function responsible for managing UE authentication and mobility registration procedures with the core network

B. Data-plane anchor that handles packet routing and forwarding between the RAN and external data networks such as the internet ✓

C. Radio access network node responsible for scheduling downlink transmissions over the air interface

D. National spectrum management entity responsible for assigning frequency bands to individual operators

Answer: B. Data-plane anchor that handles packet routing and forwarding between the RAN and external data networks such as the internet

87. The NG interface in 5G connects the gNB to the 5G core network, with:

A. NG-C (N2) carrying control-plane signalling to the AMF and NG-U (N3) carrying user-plane traffic to the UPF ✓

B. NG-C (N2) carrying user-plane traffic while NG-U (N3) carries the control-plane signalling, the reverse of the actual split

C. Both NG-C and NG-U providing a direct connection from the gNB straight out to the public internet

D. The NG interface linking two neighbouring gNBs together for direct handover coordination

Answer: A. NG-C (N2) carrying control-plane signalling to the AMF and NG-U (N3) carrying user-plane traffic to the UPF

88. The Xn interface in 5G connects:

A. The gNB directly to the 5G core network's AMF, duplicating the role of the NG-C interface

B. The UE to the gNB over the Uu air interface, rather than a link between two network nodes

C. The UPF to external data networks such as the internet, via the N6 reference point defined in the 5G core architecture

D. Neighbouring gNBs to each other, enabling handover coordination, dual connectivity and interference management ✓

Answer: D. Neighbouring gNBs to each other, enabling handover coordination, dual connectivity and interference management

89. Phased-array antennas are used in LEO satellite user terminals because they:

A. Are generally cheaper to manufacture than an equivalent mechanically steered parabolic dish antenna

B. Electronically steer narrow beams toward the moving satellite without mechanical movement, enabling rapid tracking at low latency ✓

C. Radiate omnidirectionally and therefore need essentially no beam steering or pointing mechanism toward the satellite

D. Are restricted to stationary, fixed-site installations and cannot be mounted on moving vehicles or aircraft

Answer: B. Electronically steer narrow beams toward the moving satellite without mechanical movement, enabling rapid tracking at low latency

90. Massive MIMO in 5G NR uses a large number of antenna elements at the gNB to:

A. Deliberately reduce the antenna count compared with 4G LTE base stations, simplifying RF hardware

B. Serve each UE within the cell from one single broad beam, which is assumed to simplify scheduling

C. Operate solely at sub-1 GHz frequencies, where the comparatively large wavelength is assumed to better suit big antenna arrays

D. Form narrow, high-gain beams directed at individual UEs, greatly increasing spectral efficiency through spatial multiplexing ✓

Answer: D. Form narrow, high-gain beams directed at individual UEs, greatly increasing spectral efficiency through spatial multiplexing

91. Null-steering in multi-beam satellite antennas suppresses interfering signals by:

A. Placing deep nulls in the antenna radiation pattern in the direction of interferers while maintaining high gain toward intended users ✓

B. Reducing the satellite's overall transmit power across each beam simultaneously to lower interference

C. Assigning a distinct orthogonal frequency to each individual beam at the same time
D. Physically rotating the satellite's attitude to point its beams away from interfering sources on the ground

Answer: A. Placing deep nulls in the antenna radiation pattern in the direction of interferers while maintaining high gain toward intended users

92. Higher-order modulation (e.g., 64QAM vs QPSK) transmits more bits per symbol but requires a higher SNR. In satellite systems, higher-order modulation is used when:

A. The link has a large margin above the minimum threshold, typically in good weather or for large dish terminals ✓
B. The link sits close to its minimum SNR threshold, leaving very little margin to spare for extra bits per symbol
C. Rain fade has already reduced the received signal level, which is assumed to call for packing in more bits per symbol
D. Higher-order modulation schemes are assumed to operate at a fundamentally lower SNR requirement than QPSK

Answer: A. The link has a large margin above the minimum threshold, typically in good weather or for large dish terminals

93. Adaptive Coding and Modulation (ACM) in a satellite system monitors the link quality and dynamically selects the MCS to:

A. Fix the modulation and coding scheme for the entire lifetime of a session, avoiding any signalling overhead
B. Select the highest-order modulation available at each scheduling opportunity, irrespective of current link conditions
C. Reduce the symbol rate during rain fade while deliberately holding the modulation order constant
D. Maximise throughput when the link is good and switch to more robust coding during rain fade or shadowing ✓

Answer: D. Maximise throughput when the link is good and switch to more robust coding during rain fade or shadowing

94. The carrier-to-noise density ratio C/N_0 at a satellite receiver is determined by combining the transmitter EIRP, free-space path loss, atmospheric losses and the receiver:

A. Transmit power and antenna gain considered separately rather than combined into a single receiver figure
B. The measured bit error rate together with the required Eb/N_0 threshold for that error rate
C. The transponder's total bandwidth and the number of carriers currently active within it
D. Figure of merit G/T (antenna gain divided by system noise temperature) ✓

Answer: D. Figure of merit G/T (antenna gain divided by system noise temperature)

95. The satellite power budget must carefully balance onboard power generation against consumption because:

A. Most communications satellites rely on onboard nuclear reactors, which supply effectively unlimited power to the payload
B. High-power amplifiers are assumed to consume a negligible fraction of power compared with the onboard processor
C. Solar panel area is constrained by mass and structural limits, and the high-power amplifiers (HPAs) are the dominant power consumers on the payload ✓
D. Solar panel technology is assumed to be mature enough that available power is rarely a limiting factor in modern payload design

Answer: C. Solar panel area is constrained by mass and structural limits, and the high-power amplifiers (HPAs) are the dominant power consumers on the payload

96. In a hub-and-spoke satellite network, remote VSAT terminals communicate via the satellite to a central hub, which:

A. Is physically located in orbit as part of the satellite's own payload rather than on the ground
B. Connects the satellite network to the terrestrial internet or PSTN and provides high-power, high-bandwidth uplinking to the satellite ✓

C. Has been replaced by full-mesh inter-satellite-link topologies across most modern commercial satellite systems
D. Needs no uplink of its own since it is assumed to solely receive traffic coming down from the satellite

Answer: B. Connects the satellite network to the terrestrial internet or PSTN and provides high-power, high-bandwidth uplinking to the satellite

97. A full-mesh satellite network enabled by ISLs or multi-spot beams allows:

A. Each communication path to still be routed back through a single central hub, regardless of any ISLs present
B. Terminals to bypass the satellite altogether and exchange traffic through direct terrestrial radio links
C. Terminals to communicate directly with each other through the satellite without routing through a terrestrial hub, reducing latency for terminal-to-terminal traffic ✓
D. Full-mesh connectivity to be achievable solely with GEO satellites, a capability assumed structurally unavailable to any LEO constellation

Answer: C. Terminals to communicate directly with each other through the satellite without routing through a terrestrial hub, reducing latency for terminal-to-terminal traffic

98. A satellite link availability specification of 99.9% per year means the link is unavailable for approximately:

A. About 1 second of downtime per year, a figure so small it would require just a negligible link margin
B. About 8.8 hours per year, so the link margin must be sized to withstand rain fades for all but that time ✓
C. About 36.5 days of downtime per year, which would demand an unrealistically large link margin to correct
D. Availability specifications and required link margin are treated as unrelated, independent design parameters

Answer: B. About 8.8 hours per year, so the link margin must be sized to withstand rain fades for all but that time

99. Most satellite systems use Frequency Division Duplex (FDD) rather than Time Division Duplex (TDD) because:

A. The uplink and downlink already use different frequency bands (e.g., 14 GHz up / 11 GHz down in Ku band), naturally accommodating simultaneous two-way transmission ✓
B. TDD is assumed to require two separate satellite transponders while FDD needs just a single shared transponder
C. TDD is assumed to be prohibited outright by ITU Radio Regulations for use in any commercial satellite communication system
D. FDD is assumed to require simpler transponder hardware overall than an equivalent TDD-based satellite payload

Answer: A. The uplink and downlink already use different frequency bands (e.g., 14 GHz up / 11 GHz down in Ku band), naturally accommodating simultaneous two-way transmission

100. In 5G NR, PDSCH scheduling is performed by the gNB using:

A. A fixed, static resource assignment configured once at network deployment and left unchanged thereafter
B. Uplink control signalling that the UE itself sends to reserve its own downlink resource blocks
C. Random access preambles transmitted on the PRACH, which are assumed to directly initiate PDSCH scheduling without any DCI grant
D. DCI (Downlink Control Information) transmitted on the PDCCH, which tells the UE which resource blocks carry its data in each slot ✓

Answer: D. DCI (Downlink Control Information) transmitted on the PDCCH, which tells the UE which resource blocks carry its data in each slot